Applied NLP

DATA 641

Lecture 2 — NLP Pipeline; Text Extraction and Cleanup

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly.

Text Extraction and Cleanup

Process of extracting raw text from the input data by removing all other non-textual information.

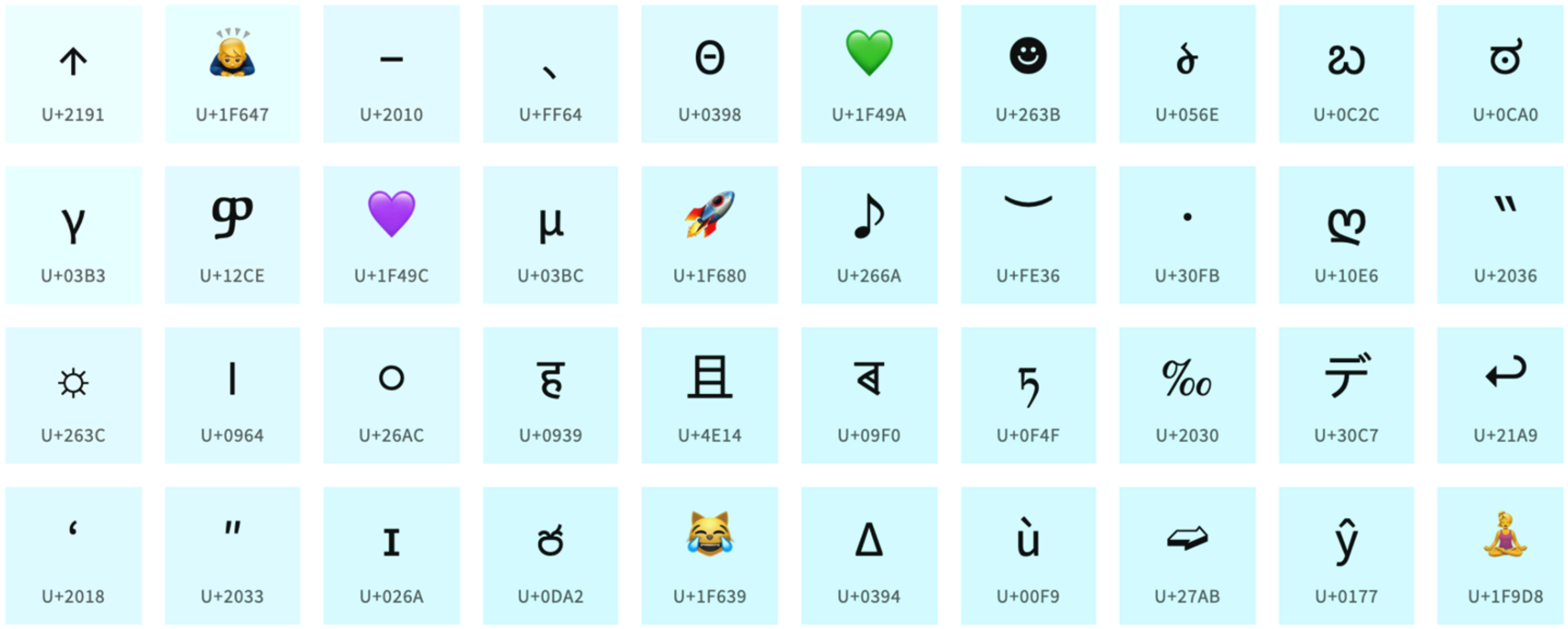
- It is an important step that has implications for all other aspects in the NLP pipeline!
- It can be one of the most time-consuming parts in an NLP project. Also, it is application dependent!

HTML Parsing and Cleanup: Extract parts of text from HTML pages

How can we extract text from an HTML page? If we observe the HTML markup of a typical Stack Overflow question page, we notice that questions and answers have special tags associated with them. We can utilize this information while extracting text from the HTML page.

- While it may seem like writing our own HTML parser is the way to go, for most cases we encounter, it is more feasible to utilize existing libraries such as Beautiful Soup and Scrapy, which provide a range of utilities to parse web pages.

Unicode Normalization: To parse non-textual symbols such as emojis and graphical characters we use Unicode normalization



Spelling Correction: This process is far from perfect! You can build your own spell checker using a huge dictionary of words from a specific language

System-Specific Error Correction:

- Extract plain text from PDF documents. While there are several libraries, such as PyPDF and PDFMiner, to extract text from PDF documents, they are far from perfect, and it's not uncommon to encounter PDF documents that can't be processed by such libraries.
- Extract plain text from scanned documents